

M4C 2025 - Assignment 1

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November 7, 2025

Problem 1

If you want some introduction to the whole problem, you can write it here ...

- (a) encrypted text: XEFVRXEFWPDJAFPCCBHBRVMDUJ
decryption process: To get back the plain text, you need to use the same 'lengthened' key but now you subtract the number of the letter in the key from the corresponding number of the letter in the encrypted text.
- (b) Let $p_1, p_2, \dots, p_n$ represent letters in P, $k_1, k_2, \dots, k_n$ letters in 'lengthened' K and $c_1, c_2, \dots, c_n$ letters in C.
Encryption: $c_i = (p_i + k_i) \bmod 26$
Decryption: $p_i = (c_i - k_i) \bmod 26$
- (c) No matter which 2 out of the 3 items are captured by the adversary, since you always catch at least a key or plain text. Knowing at least one of them can lead to knowing the other.
Captured P and K: encryption
Captured P and C: you can write down the differences between letters and derive the lengthened key. From there you just take the first iteration.
Captured K and C: decryption
- (d) We can use this table for both encrypting and decrypting process. We can imagine that the first row (or symmetrically the first column) represents the key and the first column (or symmetrically the first row) the plain text. For every letter in the plain text we can lookup its corresponding letter in the cipher text by taking the letter in the row of the letter (in plain text) and column of the letter from the key (in corresponding place). We can also use this table for decryption process by again imagining the first row/column as plain text and first column/row as key. Since we only know the cipher text and key this time, for every letter in cipher text we need to first find the letter that corresponds to it in key. We then look into that (from the key) letter's row until we find the letter from the cipher text that responds to it. The column in which this letter is situated belongs to the letter in the plain text.

Problem 2

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- (a) ...